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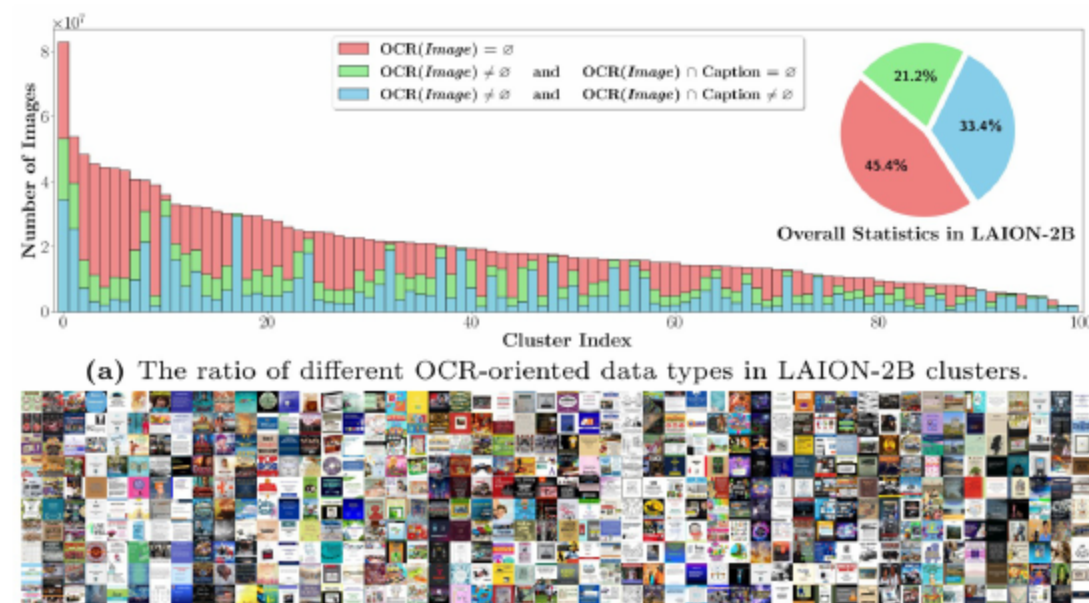
Contrastive image-text pretraining enables strong zero-shot transfer but shows biases such as reliance on embedded text, color, and gender cues. We profile LAION-2B and two model families to study text-spotting bias, connect to OCR-aware curation, and position our analysis among efforts that either leverage or mitigate visual text content.

- Observation: CLIP pipelines often over-rely on visual text present in images
- LAION-2B top-score pairs are frequently dominated by printed words in the image
- We define **Parrot Captions**: captions that repeat words printed in the image
- **Risk**: score-based filtering can amplify this text-spotting shortcut

WTF fun fact • 7625 There's a restaurant in the base of a waterfall in Philippines.	fact • 7625 There's a restaurant base of a	fact • 7625 There's a restaurant base of a	WTF fun fact • 7625 There's a restaurant in the base of a waterfall in Philippines.	You
Raw Image	All-Emb. Text Removal	Co-Emb. Text Removal	Random Inpainted	Syn-Emb. Text Template
Waterfall restaurants in the Philippines - WTF fun facts				Architectura
Caption	All-Emb. Text Mask	Co-Emb. Text Mask	Random Sampled Mask	Syn-Emb. Text Template

- Definitions: **Embedded text** are OCR-recognized words in pixels; **Co-embedded text** are OCR words also present in the caption
- Processing: CLIP-feature clustering, OCR recognition, and caption-OCR matching with exact and fuzzy rules
- Counterfactuals: inpainting-based removal of all embedded words or only co-embedded words, plus random-mask controls

04 Clustering, OCR, and Matching at Scale



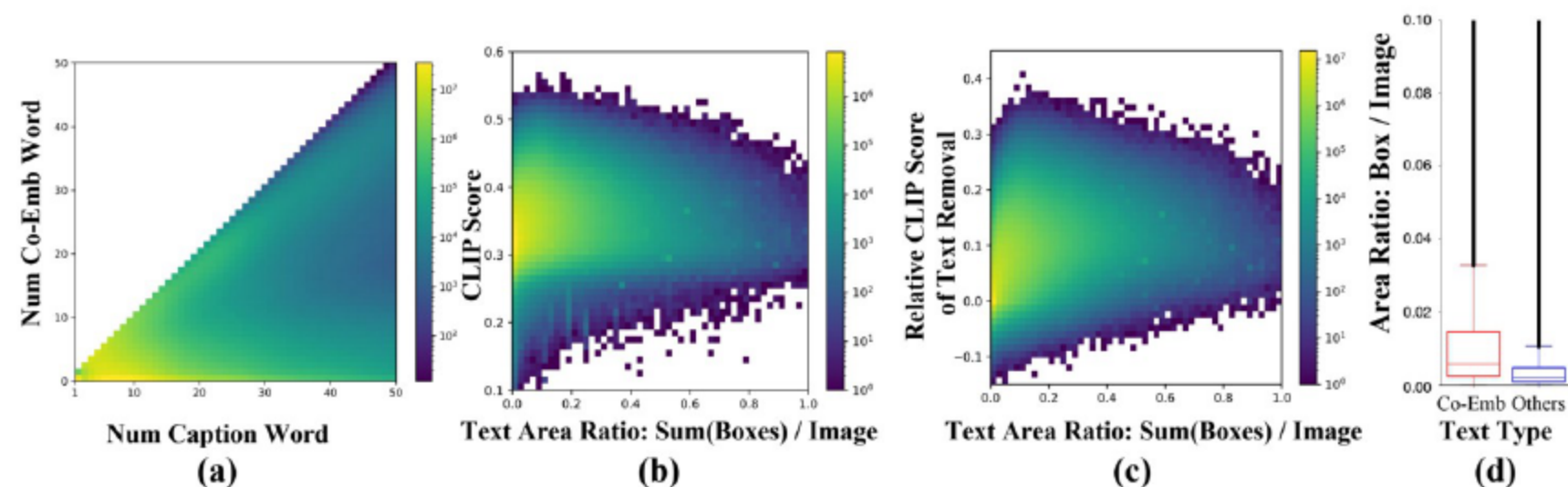
- Built 4,000 CLIP-feature clusters with dimensionality reduction and ran OCR over all images
- Grouped pairs into: no text, text without overlap, and text with co-embedded words
- Top-score samples are dominated by posters, books, ads, and slides rich in text
- Finding:** high-similarity pairs often hinge on overlapping words

05 Overall Statistics

Number of Total Images	1,985,284,122
Number of Images w/ Emb. Text	1,083,896,427
Number of Images w/ Co-Emb. Text	663,600,432
Co-Emb. Text Rate (in Total)	15.42%
Co-Emb. Text Rate (in Images w/ Emb. Text)	28.24%
Fuzzy Co-Emb. Text Rate (in Total)	30.46%
Fuzzy Co-Emb. Text Rate (in Images w/ Emb. Text)	55.79%

- About half of images contain OCR-detected embedded text
- Around 15 percent of caption words are co-embedded, rising toward 30 percent with fuzzy matching
- Parrot-style overlap is prevalent across many visual clusters
- Implication:** training data strongly exposes models to text spotting

06 Co-Emb. Text Influence on CLIP Scores



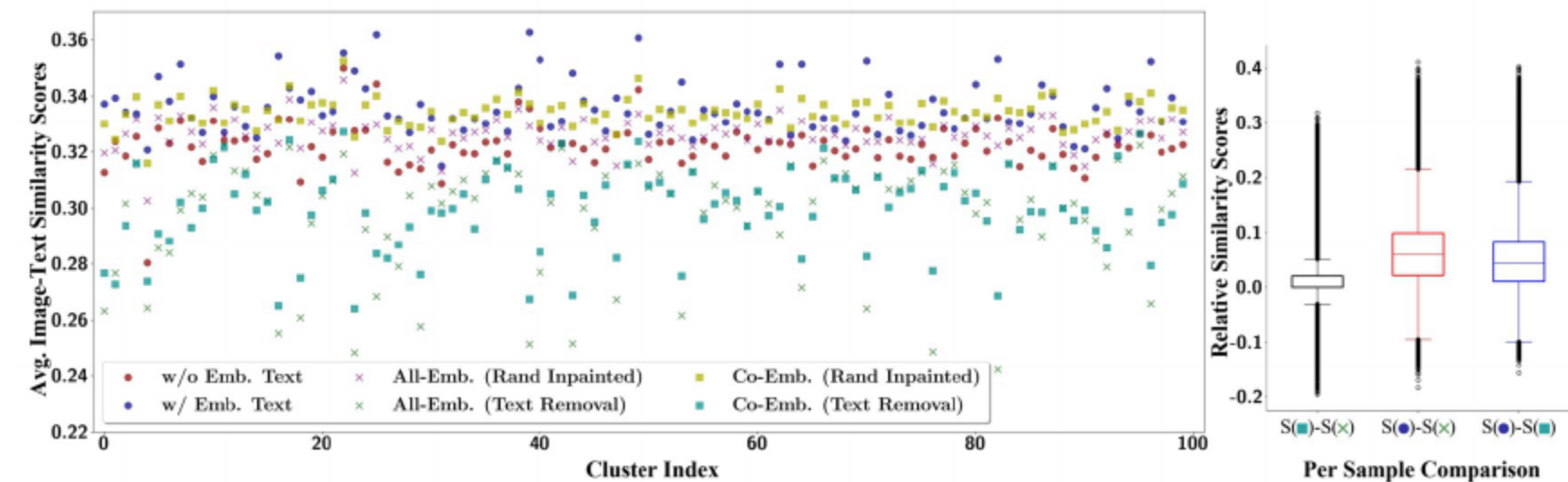
- Even small co-embedded text can drive similarity; removing it lowers scores
- Text size alone is not predictive; content and resolution matter
- Larger printed words are more likely to be repeated in captions
- Sensitivity:** scores often hinge on overlapping tokens rather than semantics

07 From Dataset Bias to CLIP Filtering

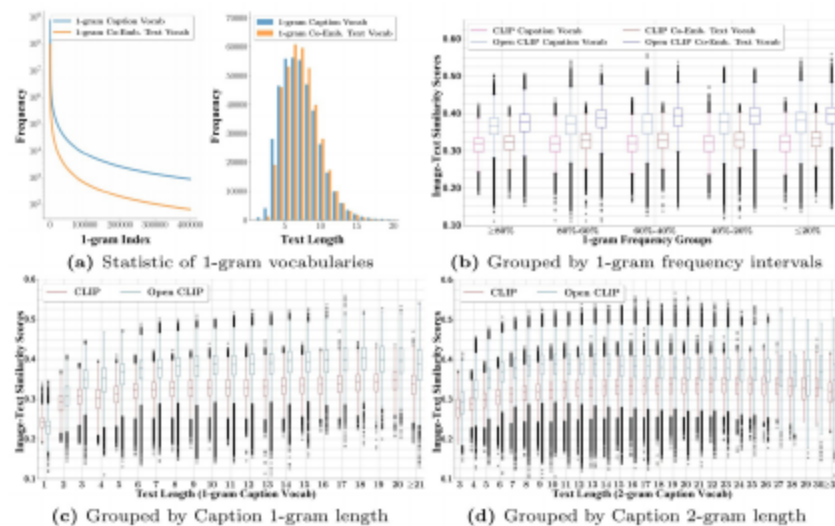
Because LAION-2B was filtered by CLIP scores, we test whether embedded or co-embedded text drives selection via text inpainting, and probe token preferences with synthetic n-grams to compare CLIP against OpenCLIP.

Setup	Average CLIP Score
● Raw Images w/o Emb. Text	0.3223 ± 0.0078
● Raw Images w/ Emb. Text	0.3358 ± 0.0094
× All-Emb. Random Inpainted	0.3260 ± 0.0057
× All-Emb. Text Removal	0.2974 ± 0.0197
■ Co-Emb. Random Inpainted	0.3341 ± 0.0051
■ Co-Emb. Text Removal	0.2993 ± 0.0146

- Apply fast marching inpainting on OCR polygons for all embedded text and for co-embedded words only
- Use random inpainting masks from other images as controls
- Compare CLIP scores across raw, random-inpaint, and text-removed variants
- **Control strength:** random masks verify robustness to inpainting artifacts



- Images with embedded text score higher than images without text
- Removing text reduces scores beyond random-inpaint controls
- Some samples improve after removal, showing text can distract
- **Conclusion:** overlapping words systematically inflate similarity



- OpenCLIP exhibits stronger text spotting than CLIP
- Both models are more sensitive to vocabularies derived from co-embedded words
- Longer tokens are easier to spot; very long bigrams degrade
- **Bias signal:** training lineage correlates with stronger spotting

We train CLIP and BLIP on subsets stratified by embedded text presence, co-embedded text rate, and relative score after text removal. Evaluation covers zero-shot classification and retrieval, synthetic probes, and a real validation split with and without embedded text.

12 Embedded Text Presence Harms Generalization

- Training only on images with embedded text reduces zero-shot performance
- Removing embedded-text images improves average, ImageNet, and retrieval tracks
- **Gain:** stronger text-reading ability on synthetic probes
- **Trade-off:** improved reading but weaker general vision alignment

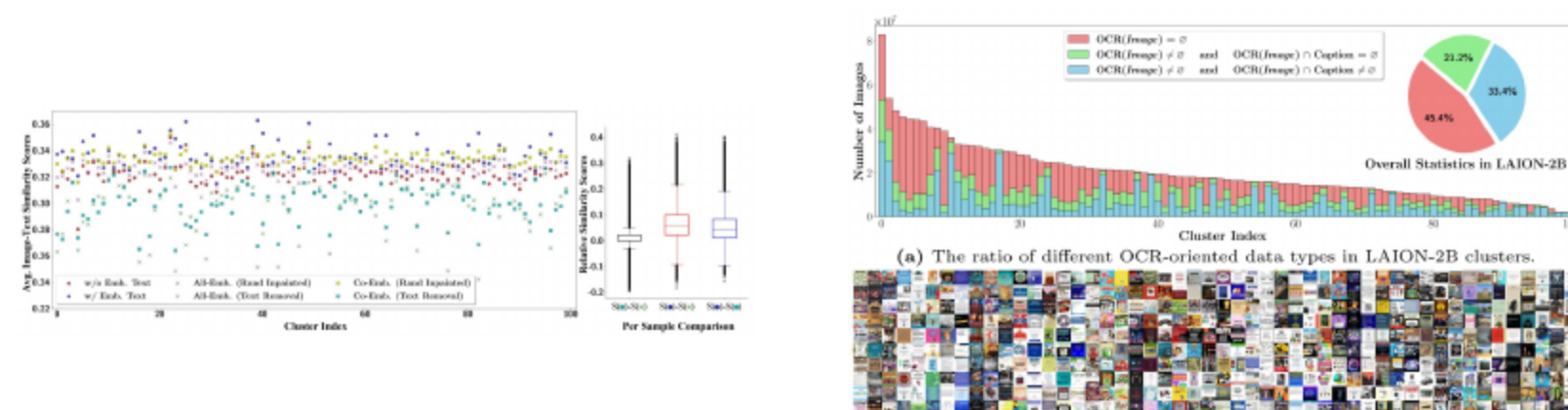
Data	Model	IN	Ret.	Avg.
3M Random	RN50	0.204	0.222	0.256
3M w/o Emb. Text	RN50	0.228	0.232	0.282
3M w/ Emb. Text Only	RN50	0.071	0.139	0.164
3M Random	ViT-B	0.131	0.148	0.210
3M w/o Emb. Text	ViT-B	0.162	0.164	0.234
3M w/ Emb. Text Only	ViT-B	0.052	0.111	0.153
12M Random	RN50	0.360	0.354	0.354
12M w/o Emb. Text	RN50	0.409	0.361	0.372
12M w/ Emb. Text Only	RN50	0.129	0.192	0.218
12M Random	ViT-B	0.314	0.299	0.351
12M w/o Emb. Text	ViT-B	0.370	0.318	0.364
12M w/ Emb. Text Only	ViT-B	0.129	0.172	0.225

13 Effect of Co-Embedded Text Rate (CoTR)

- Higher co-embedded text rate severely degrades zero-shot performance
- Co-embedded rate of zero performs on par with random despite text being present
- Overlap in words, not mere text presence, drives harm
- **Guidance:** limit overlap-heavy pairs during curation

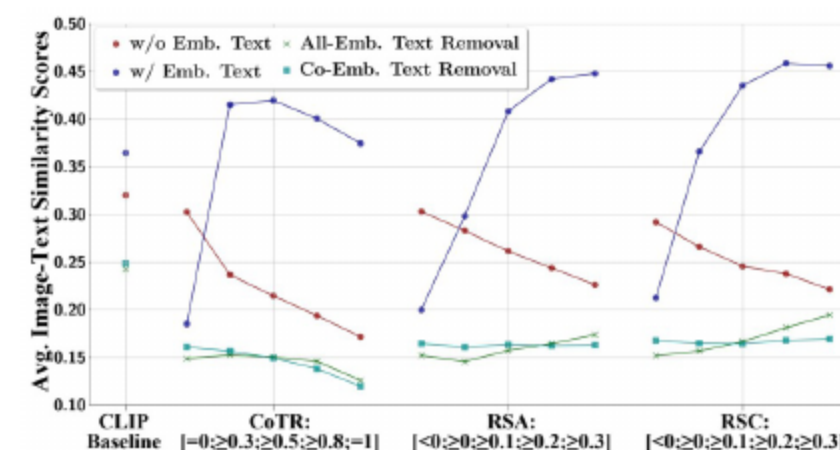
Data (3M)	Model	Avg.S(●)	IN	Ret.	Avg.
RSA < 0.0	RN50	0.319	0.181	0.220	0.239
RSA ≥ 0.0	RN50	0.339	0.126	0.180	0.215
RSA ≥ 0.1	RN50	0.351	0.041	0.123	0.148
RSA ≥ 0.2	RN50	0.360	0.017	0.094	0.109
RSA ≥ 0.3	RN50	0.376	0.009	0.075	0.097
RSA < 0.0	ViT-B	0.319	0.123	0.159	0.198
RSA ≥ 0.0	ViT-B	0.339	0.079	0.129	0.185
RSA ≥ 0.1	ViT-B	0.351	0.031	0.103	0.134
RSA ≥ 0.2	ViT-B	0.360	0.012	0.080	0.103
RSA ≥ 0.3	ViT-B	0.376	0.006	0.070	0.096

14 Relative Score Thresholding (RSA/RSC)



- Increasing relative score thresholds yields worse zero-shot results
- Raw CLIP scores rise with these thresholds, revealing filtering risk
- Samples dominated by caption overlap are least informative for training
- **Warning:** score-based selection can overfit to text spotting

15 Real-Image Text-Spotting vs Natural Images



- **Positive:** parrot-heavy training improves alignment on overlap-dominated pairs
- **Negative:** alignment worsens on images without embedded text
- Text removal further confirms sensitivity to overlapping words
- Balanced mixes are needed to retain broad generalization

BLIP Data (3M)	Visual Question Answering (Acc)			Image Captioning (CIDEr)		Text-to-Image Retrieval (R@1)		Image-to-Text Retrieval (R@1)	
	VQAv2	TextVQA	ST-VQA	COCO	TextCaps	COCO	TextCaps	COCO	TextCaps
Rand	71.07	15.36	10.48	115.6	53.7	48.91	56.34	65.46	72.45
w/ Emb. Text	68.94	19.05	12.65	108.9	92.1	42.89	70.1	58.5	81.42
w/o Emb. Text	71.22	13.65	9.29	116.2	41.5	49.96	31.83	66.5	48.7
CoTR = 0.0	71.11	13.97	9.75	116.3	44.6	49.55	38.05	66.08	54.57
CoTR ≥ 0.3	67.4	19.28	11.81	104.9	96.9	37.78	67.28	51.98	78.2
CoTR ≥ 0.5	67.02	19.64	12.38	102.7	94.1	35.94	65.24	50.32	76.94
CoTR ≥ 0.8	66.38	18.50	12.00	100.9	91.6	34.13	62.65	46.9	73.56
CoTR = 1.0	66.18	18.47	12.80	101.2	91.3	33.55	61.83	46.62	73.05
RSA < 0.0	70.79	14.16	9.64	115.7	44.9	48.25	36.85	64.72	54.7
RSA ≥ 0.0	70.03	18.76	11.81	111.9	84.5	46.25	68.61	62.92	81.23
RSA ≥ 0.1	68.14	19.48	13.33	105.6	96.1	39.96	68.13	54.64	79.37
RSA ≥ 0.2	66.01	21.06	11.85	98.7	94.4	33.03	64.17	47.12	75.33
RSA ≥ 0.3	64.20	18.44	12.04	95.26	91.1	26.64	60.11	37.3	70.24
RSC < 0.0	70.13	15.19	10.74	112.2	46.7	46.8	41.95	63.24	58.05
RSC ≥ 0.0	68.86	20.12	13.75	107.8	93.5	42.0	69.78	57.42	80.92
RSC ≥ 0.1	67.35	20.54	12.84	103.4	96.9	36.4	66.69	51.02	77.79
RSC ≥ 0.2	62.62	20.32	13.14	98.7	92.8	30.08	61.38	42.96	71.98
RSC ≥ 0.3	63.75	18.94	13.03	92.9	88.7	24.23	58.35	34.72	68.95

- **Gain:** parrot-centric pretraining boosts text-reading tasks
- **Cost:** hurts natural-image tasks like VQA and COCO
- At high thresholds, both reading and seeing degrade
- Careful data mixing is required for balanced capabilities

LAION-2B contains extensive embedded text with caption overlap, and CLIP scores are highly sensitive to such overlap. Score-based filtering can reinforce this bias. Favor OCR-aware debiasing and balanced mixtures to preserve generalization while retaining text-reading capability.

THANKS!